DSA0601 Data Handling and Visualization
Slot A
CO2 – Assessment Test 6 (AT3)
Case Study
Instructions:

For each question, write the program in the specified language (R or Python), generate the required
visualization, and answer the interpretation sub-questions clearly with justification.

Case Study 1 – Retail Sales Performance Dashboard

Scenario

A retail company has collected one year of sales data from five regions (North, South, East, West, Central). The dataset contains monthly sales, product categories, customer segments, and total revenue.

The management wants to identify:

- The highest-performing region.
- Monthly sales trends.
- Revenue contribution of each product category.
- Relationship between advertising expenditure and sales.
- Sales distribution across regions.

Tasks

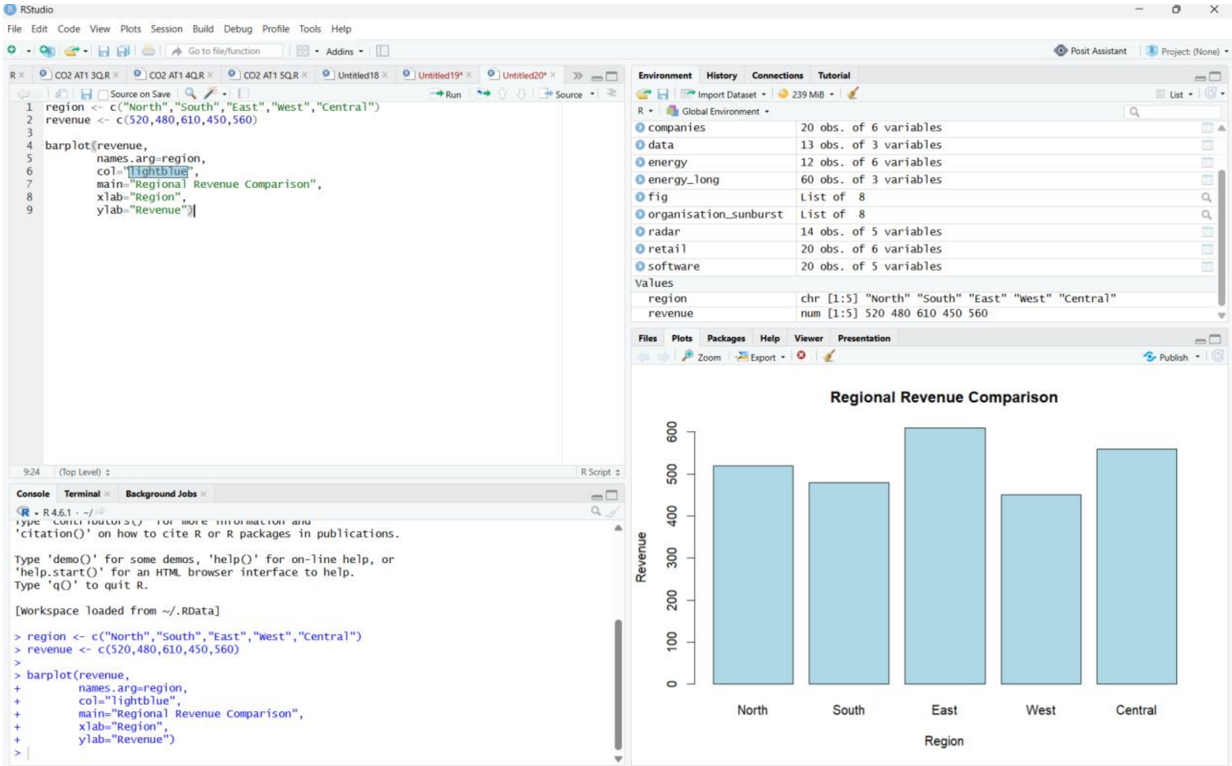
1. Select the most suitable visualization for each requirement:
    - Regional revenue comparison
    - Monthly sales trend
    - Product category contribution
    - Advertising expenditure vs sales
    - Regional sales distribution

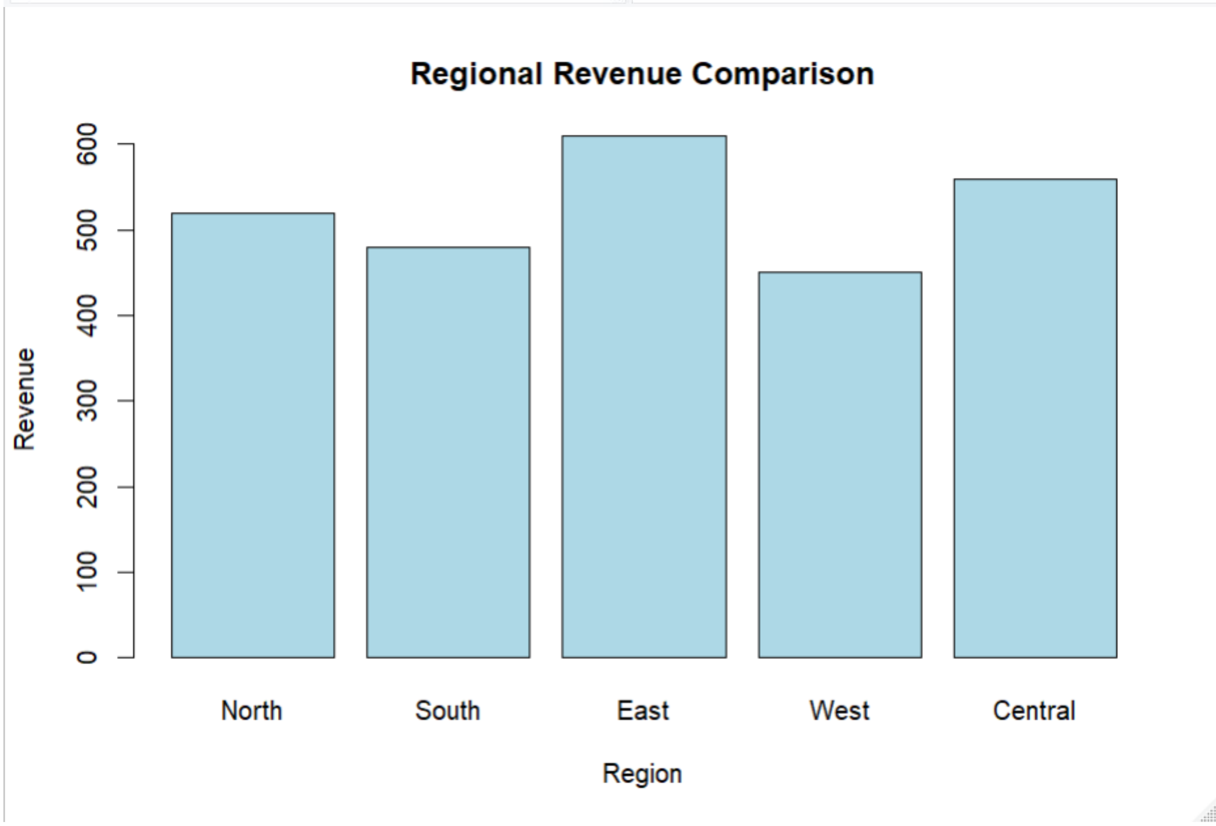
Answer:

Requirement	**Suitable Visualization**	**Justification**
Regional revenue comparison	**Bar Plot**	Compares revenue across different regions clearly.
Monthly sales trend	**Line Chart**	Displays sales changes over time effectively.
Product category contribution	**Stacked Bar Chart**	Shows the contribution of each product category to total revenue.
Advertising expenditure vs sales	**Scatter Plot**	Identifies the relationship between advertising expenditure and sales.
Regional sales distribution	**Histogram**	Displays how sales values are distributed among regions.

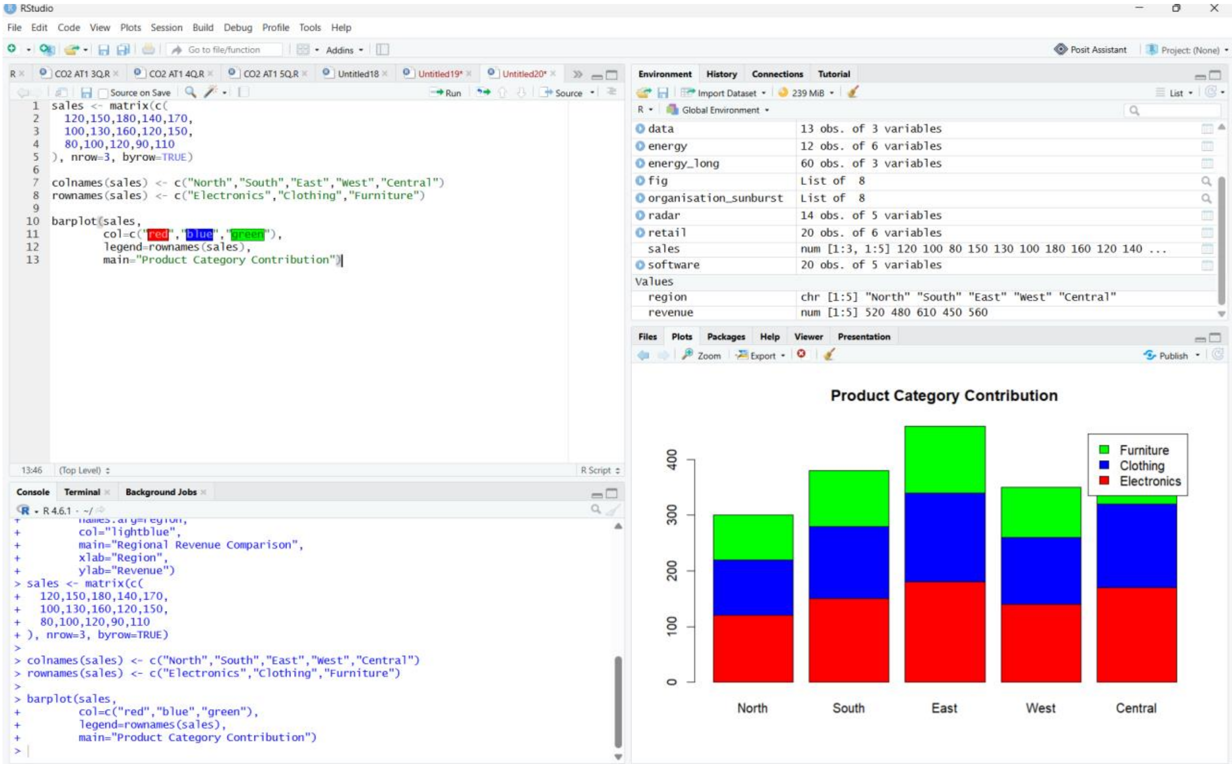
2. Using **R**, create:

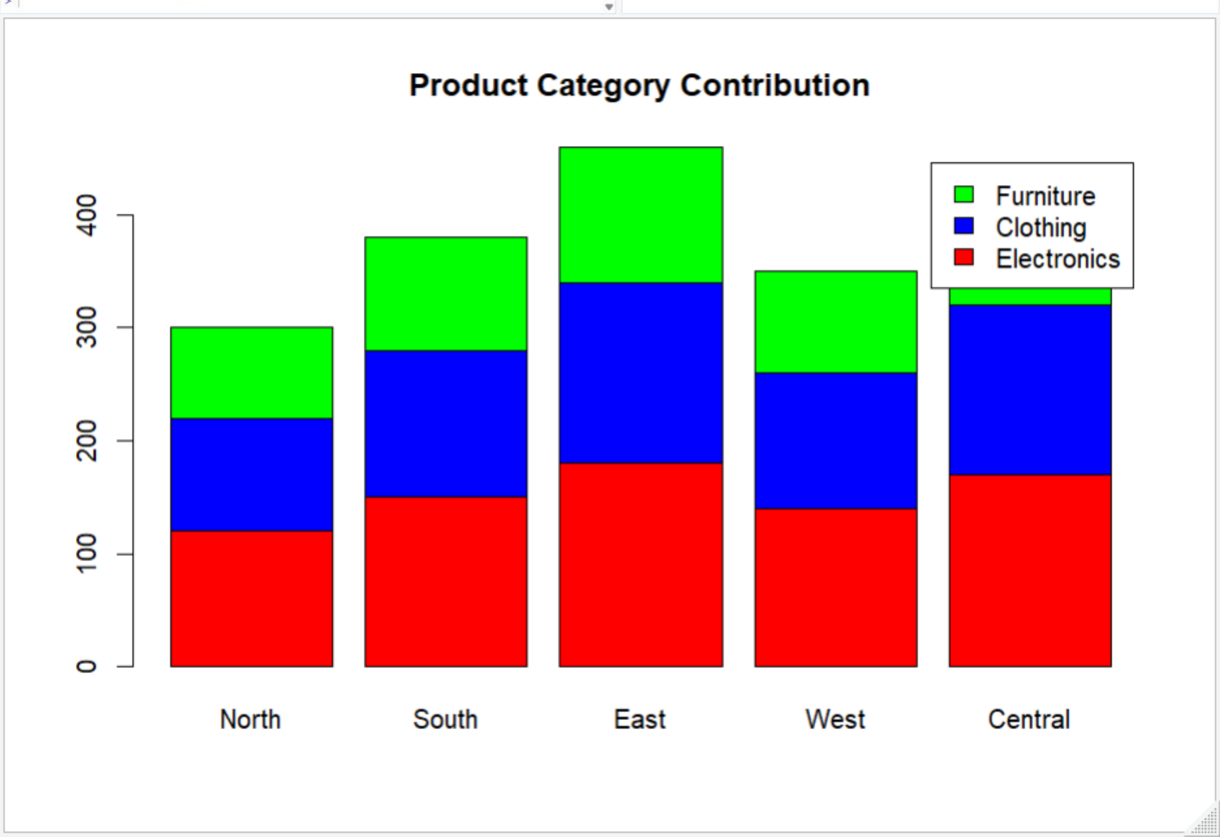
- **Bar Plot**



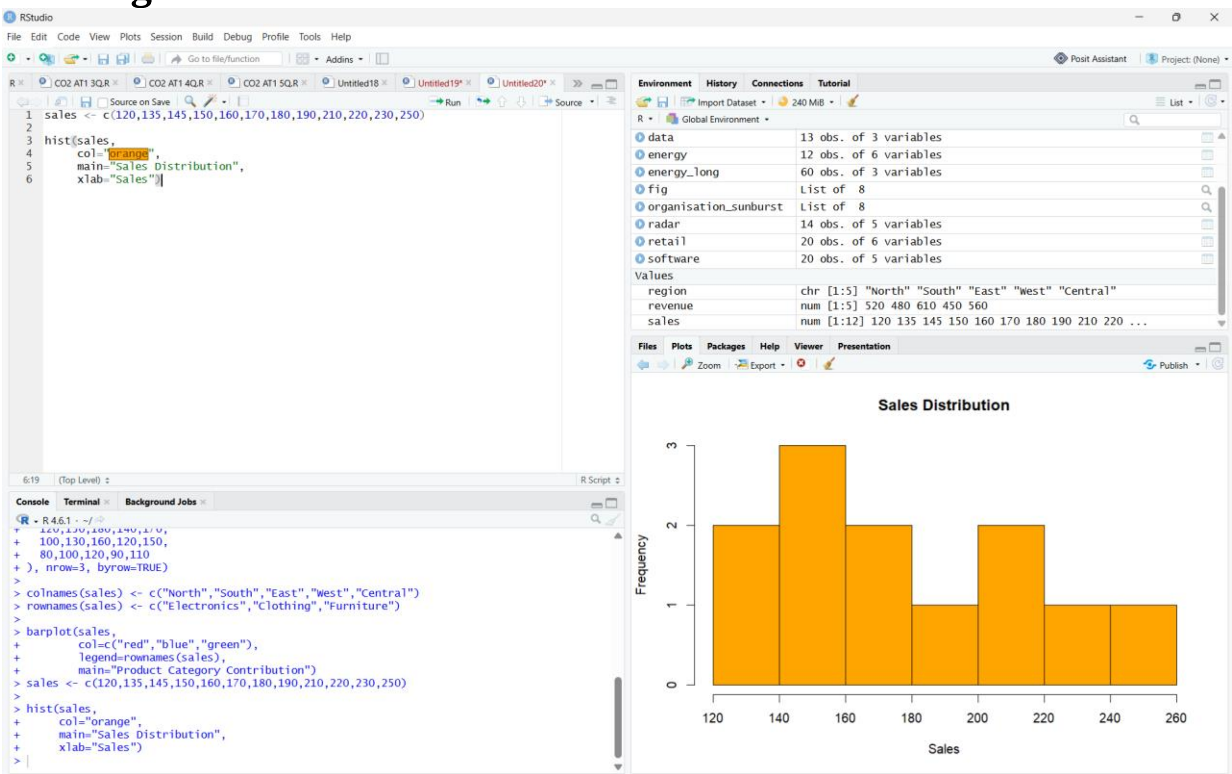


- ## Stacked Bar Chart

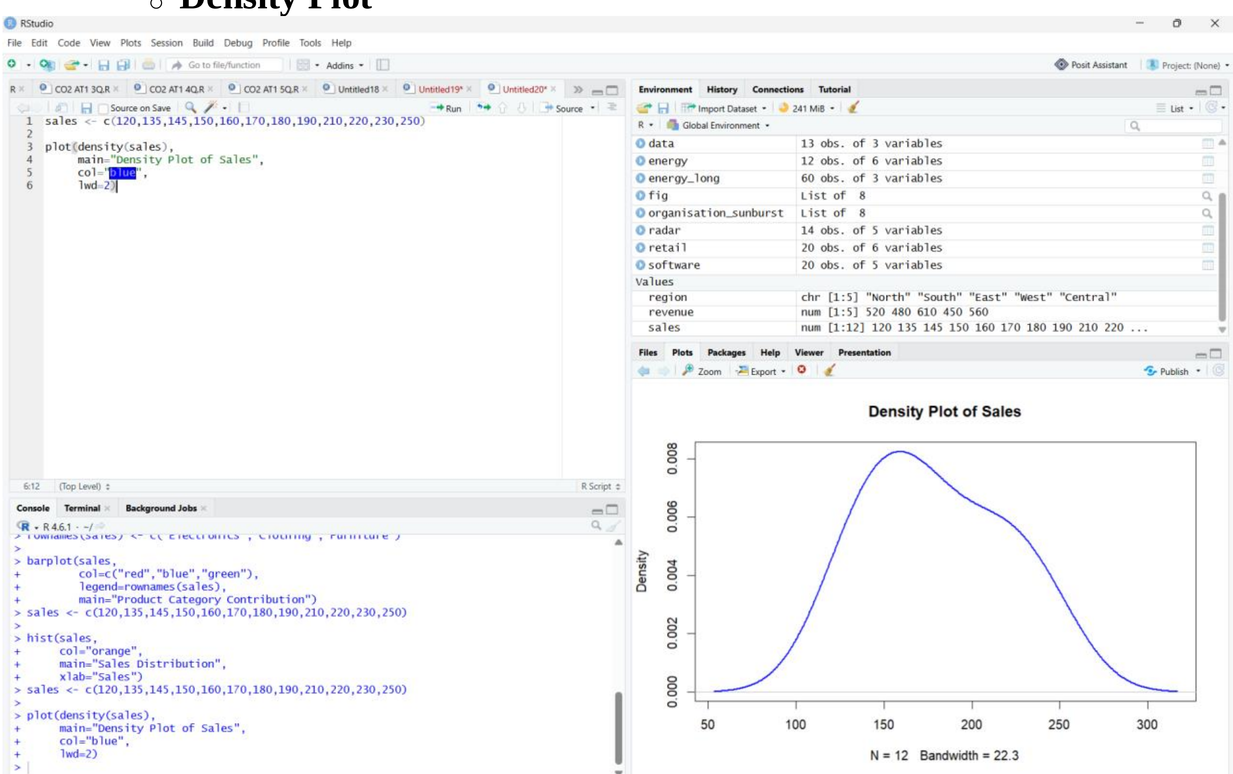




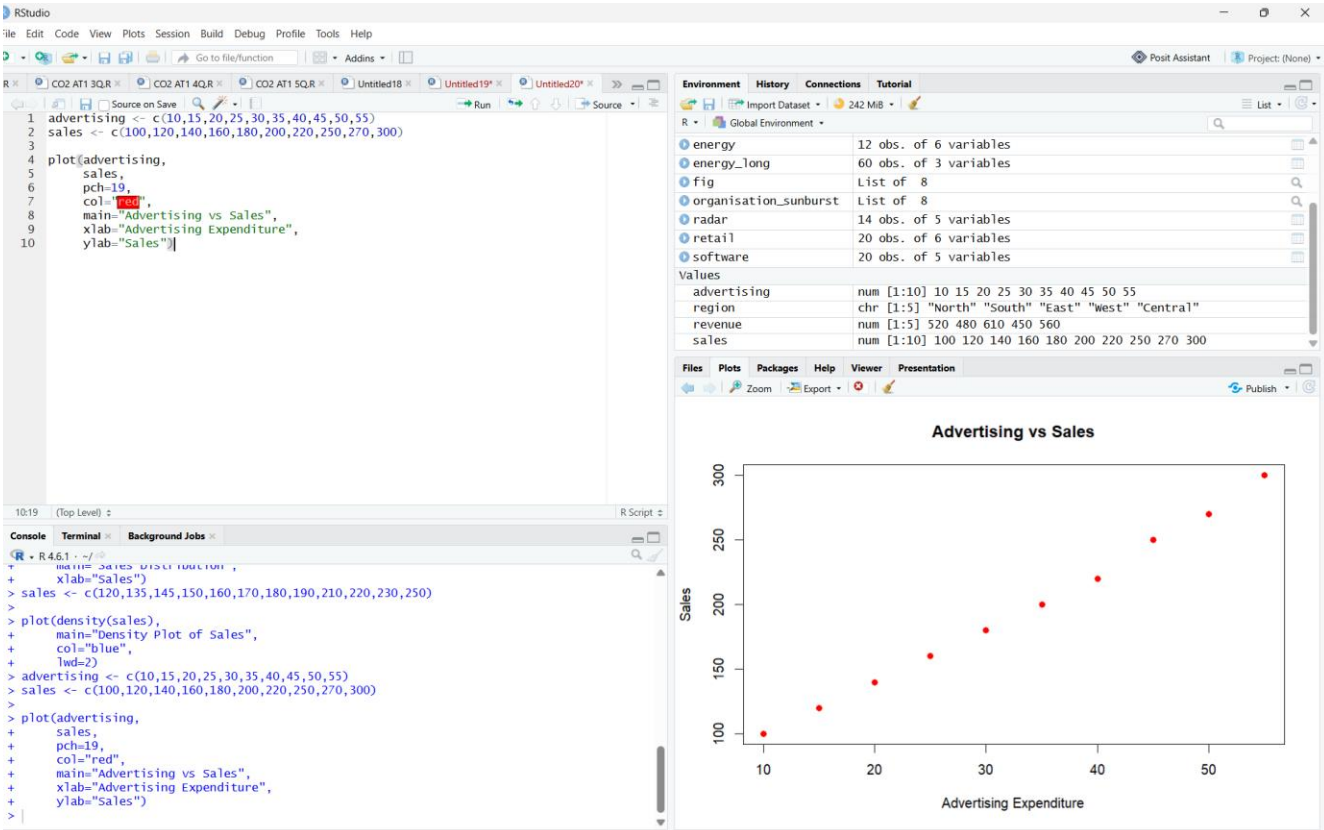
- ## Histogram



- ## Density Plot



○ **Scatter Plot**



3. Interpret:

○ Which region generated the highest revenue?

Answer:

The **East** region generated the highest revenue because it has the largest revenue value among all regions.

○ Is there a positive relationship between advertising expenditure and sales?

Answer:

Yes. The scatter plot shows a **positive relationship**, indicating that as advertising expenditure increases, sales also increase.

○ Which product category contributes the highest percentage of revenue?

Answer:

The **Electronics** category contributes the highest percentage of revenue, as it has the largest values in the stacked bar chart.

Case Study 2 – University Student Performance Analysis

Scenario

A university stores examination data for four departments:

- Computer Science

- Information Technology
- Electronics
- Mechanical

The dataset includes:

- Semester marks
- Attendance percentage
- CGPA
- Placement status

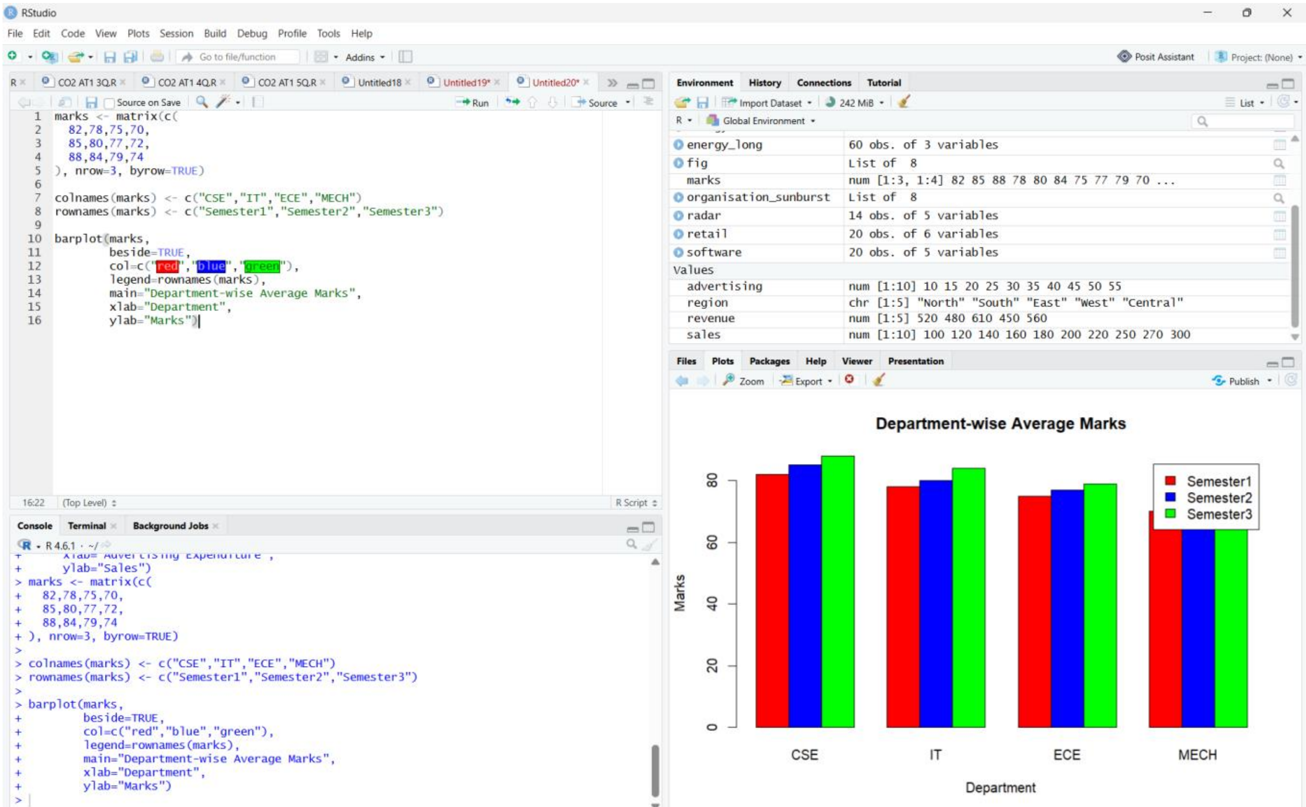
The academic committee wants to evaluate student performance and placement readiness. **Tasks**

1. Choose appropriate visualizations for:
   - Department-wise average marks
   - Distribution of CGPA
   - Placement percentage
   - Attendance vs CGPA relationship
   - Semester-wise performance trend

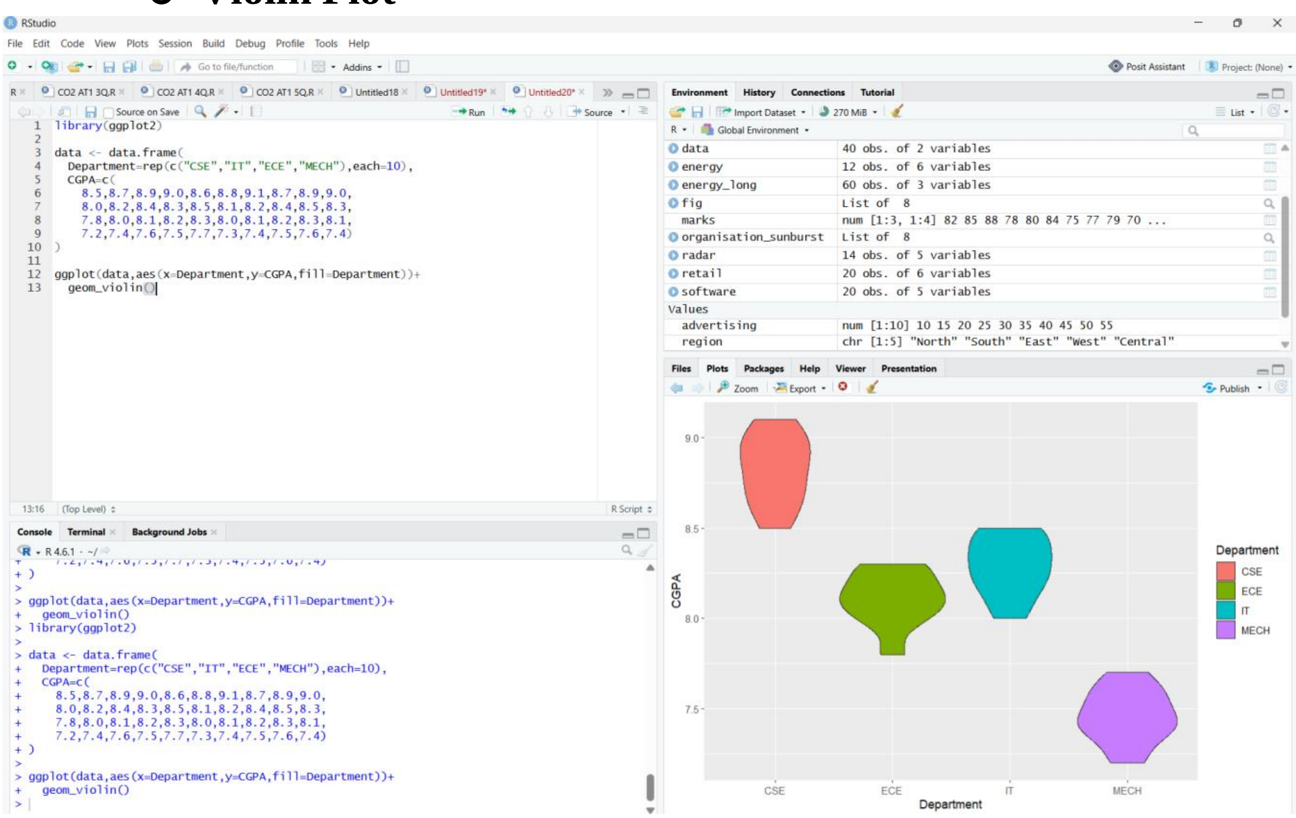
Answer:

Requirement	**Suitable Visualization**	**Justification**
Department-wise average marks	**Grouped Bar Chart**	Compares average marks across different departments.
Distribution of CGPA	**Histogram with Density Curve**	Shows the distribution and pattern of CGPA values.
Placement percentage	**Bar Chart**	Clearly compares placement percentages among departments.
Attendance vs CGPA relationship	**Scatter Plot**	Identifies the relationship between attendance and CGPA.
Semester-wise performance trend	**Line Chart**	Displays performance changes over semesters.

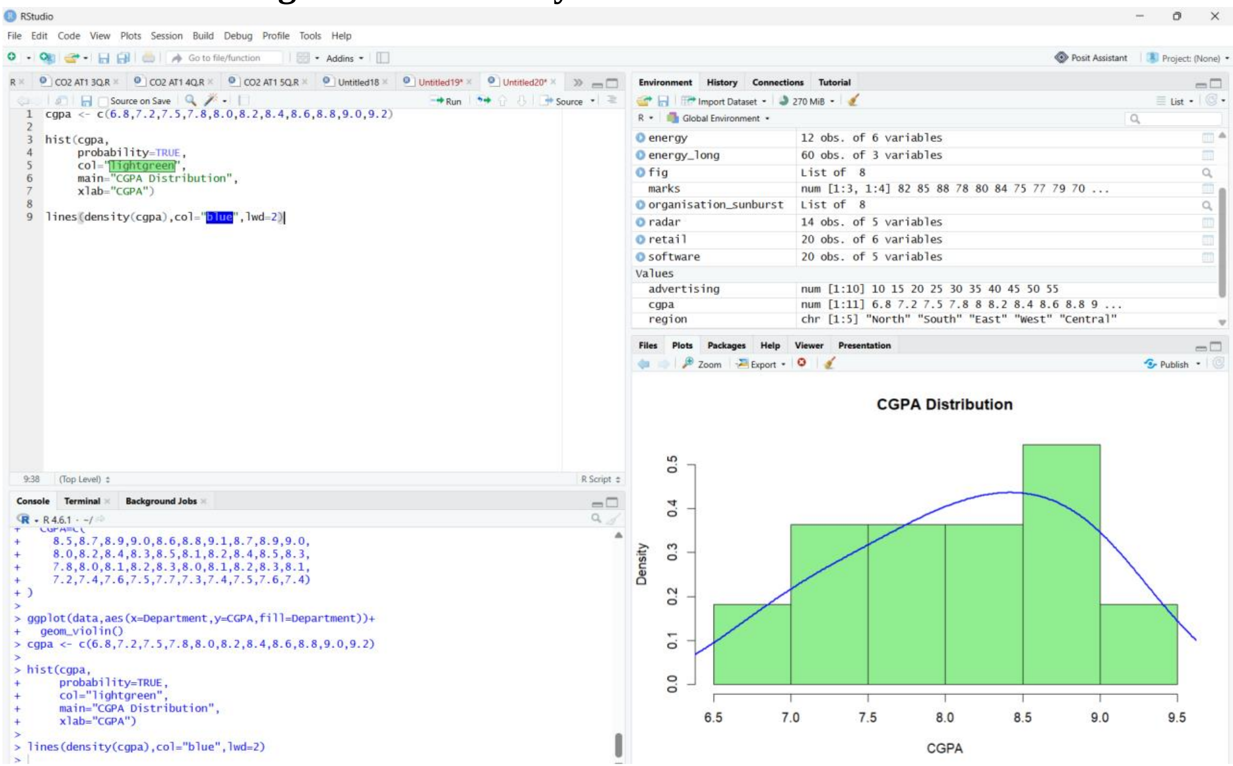
2. Develop the following in **R**:
   - **Grouped Bar Chart**


RStudio
marks <- matrix(c(
82,78,75,70,
85,80,77,72,
88,84,79,74
), nrow=3, byrow=TRUE)
colnames(marks) <- c("CSE","IT","ECE","MECH")
rownames(marks) <- c("Semester1","Semester2","Semester3")
barplot(marks,
beside=TRUE,
col=c("red","blue","green"),
legend=rownames(marks),
main="Department-wise Average Marks",
xlab="Department",
ylab="Marks")
energy_long 60 obs. of 3 variables
fig List of 8
marks num [1:3, 1:4] 82 85 88 78 80 84 75 77 79 70 ...
organisation_sunburst List of 8
radar 14 obs. of 5 variables
retail 20 obs. of 6 variables
software 20 obs. of 5 variables
Values
advertising num [1:10] 10 15 20 25 30 35 40 45 50 55
region chr [1:5] "North" "South" "East" "West" "Central"
revenue num [1:5] 520 480 610 450 560
sales num [1:10] 100 120 140 160 180 200 220 250 270 300
Department-wise Average Marks
Semester1
Semester2
Semester3
Marks
0
20
40
60
80
CSE
IT
ECE
MECH
Department

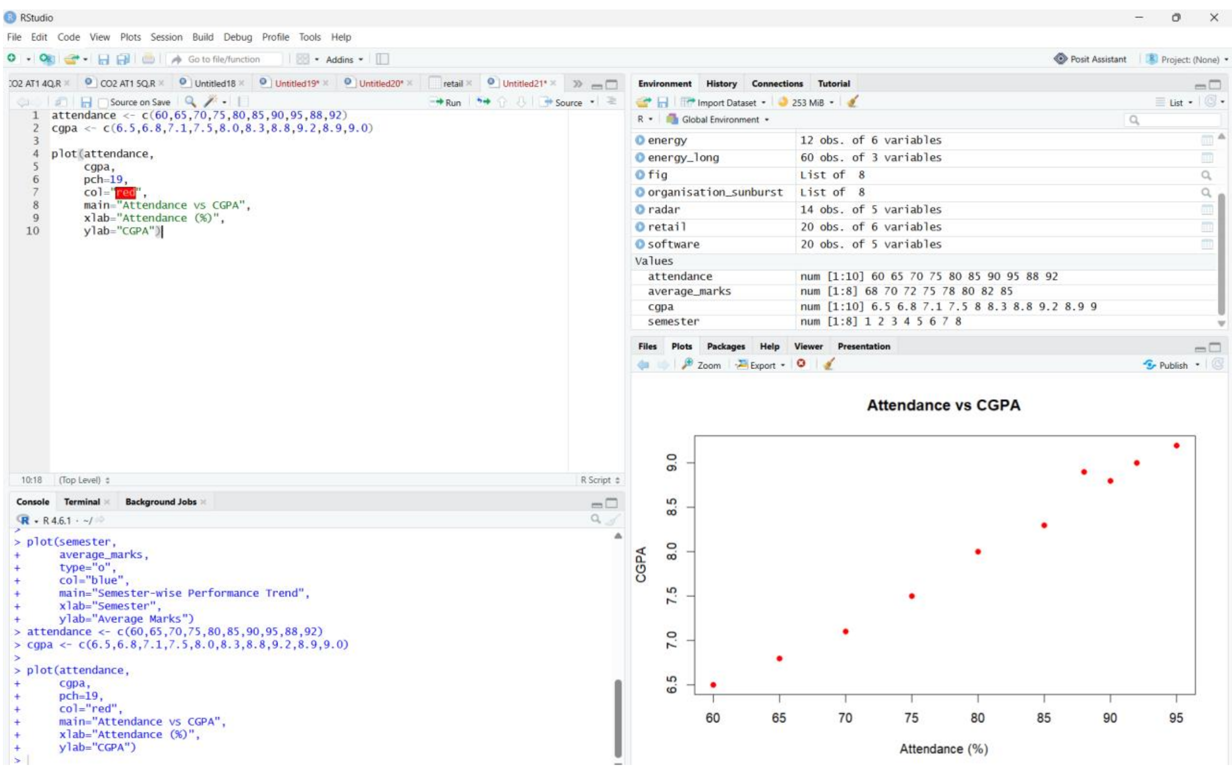
- **Violin Plot**


RStudio
library(ggplot2)
data <- data.frame(
Department=rep(c("CSE","IT","ECE","MECH"),each=10),
CGPA=c(
8.5,8.7,8.9,9.0,8.6,8.8,9.1,8.7,8.9,9.0,
8.0,8.2,8.4,8.3,8.5,8.1,8.2,8.4,8.5,8.3,
7.8,8.0,8.1,8.2,8.3,8.0,8.1,8.2,8.3,8.1,
7.2,7.4,7.6,7.5,7.7,7.3,7.4,7.5,7.6,7.4)
)
ggplot(data,aes(x=Department,y=CGPA,fill=Department))+
geom_violin()
data 40 obs. of 2 variables
energy 12 obs. of 6 variables
energy_long 60 obs. of 3 variables
fig List of 8
marks num [1:3, 1:4] 82 85 88 78 80 84 75 77 79 70 ...
organisation_sunburst List of 8
radar 14 obs. of 5 variables
retail 20 obs. of 6 variables
software 20 obs. of 5 variables
Values
advertising num [1:10] 10 15 20 25 30 35 40 45 50 55
region chr [1:5] "North" "South" "East" "West" "Central"
CGPA
9.0
8.5
8.0
7.5
CSE
ECE
IT
MECH
Department
Department
CSE
ECE
IT
MECH

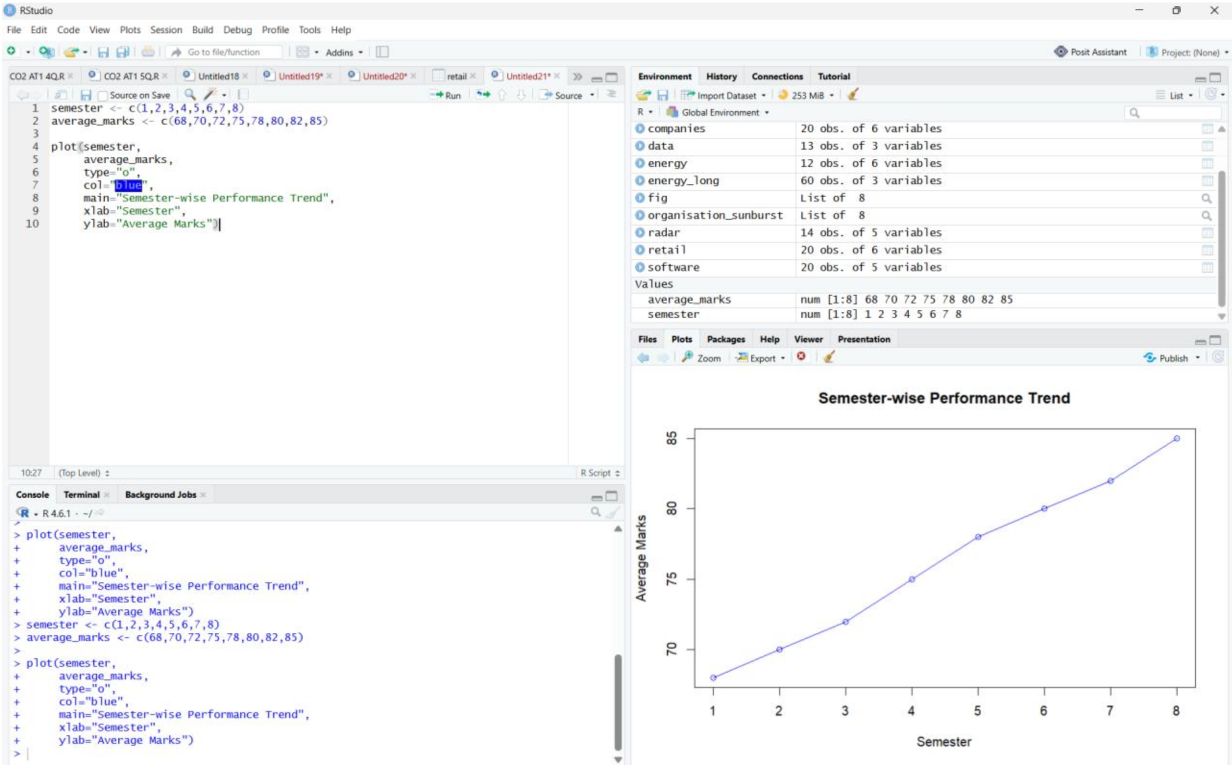
Histogram with Density Curve



Scatter Plot



- **Line Chart**



3. Interpret:

○ Which department has the highest average CGPA?

Answer:

The Computer Science (CSE) department has the highest average CGPA among all departments.

○ Does higher attendance improve CGPA?

Answer:

Yes. The scatter plot indicates a **positive relationship**, showing that students with higher attendance generally achieve higher CGPA.

○ Which department has the highest placement rate?

Answer:

The **Computer Science (CSE)** department has the highest placement rate compared to the other departments.

Case Study 3 – Hospital Patient Analytics

Scenario

A multi-specialty hospital records patient information including: • Department

- Age
- Treatment cost
- Recovery time
- Disease category
- Admission month

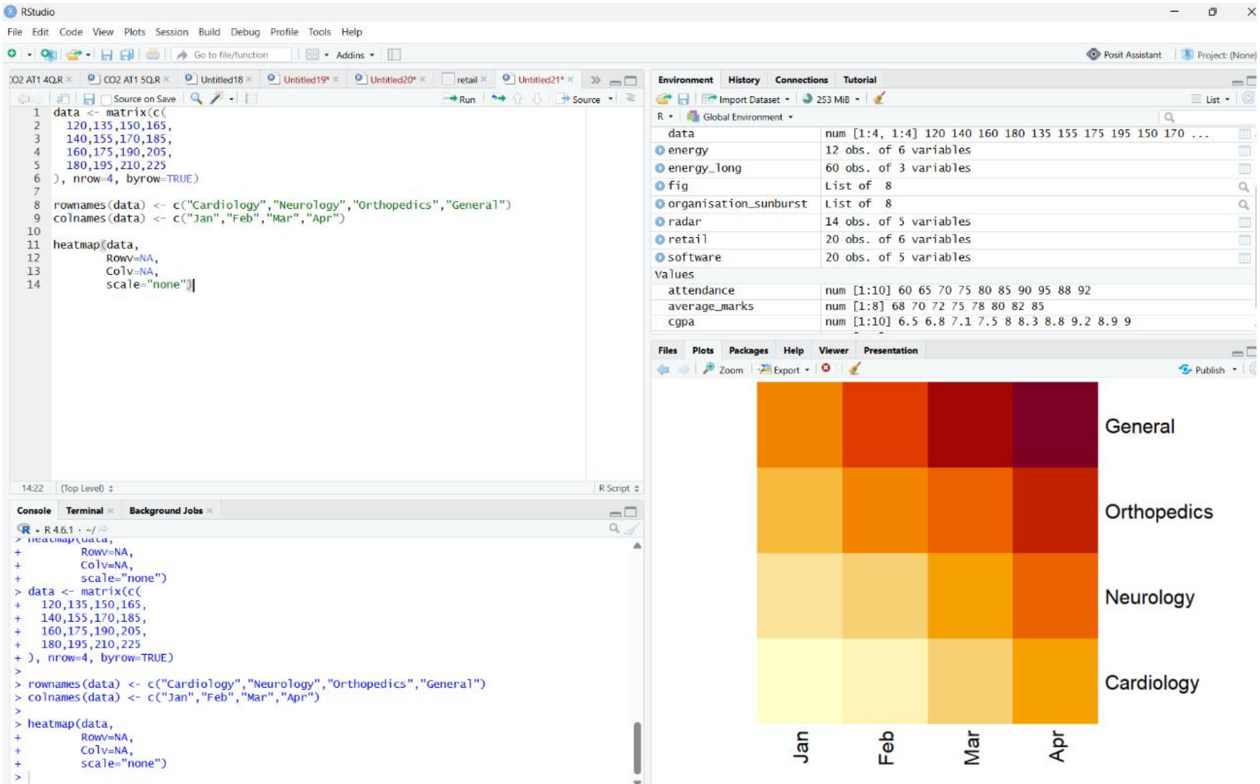
Hospital administrators want to improve resource planning. **Tasks**

1. Select suitable visualizations for:
    - Monthly patient admissions
    - Distribution of treatment costs
    - Disease category proportions
    - Age vs recovery time
    - Department-wise treatment costs

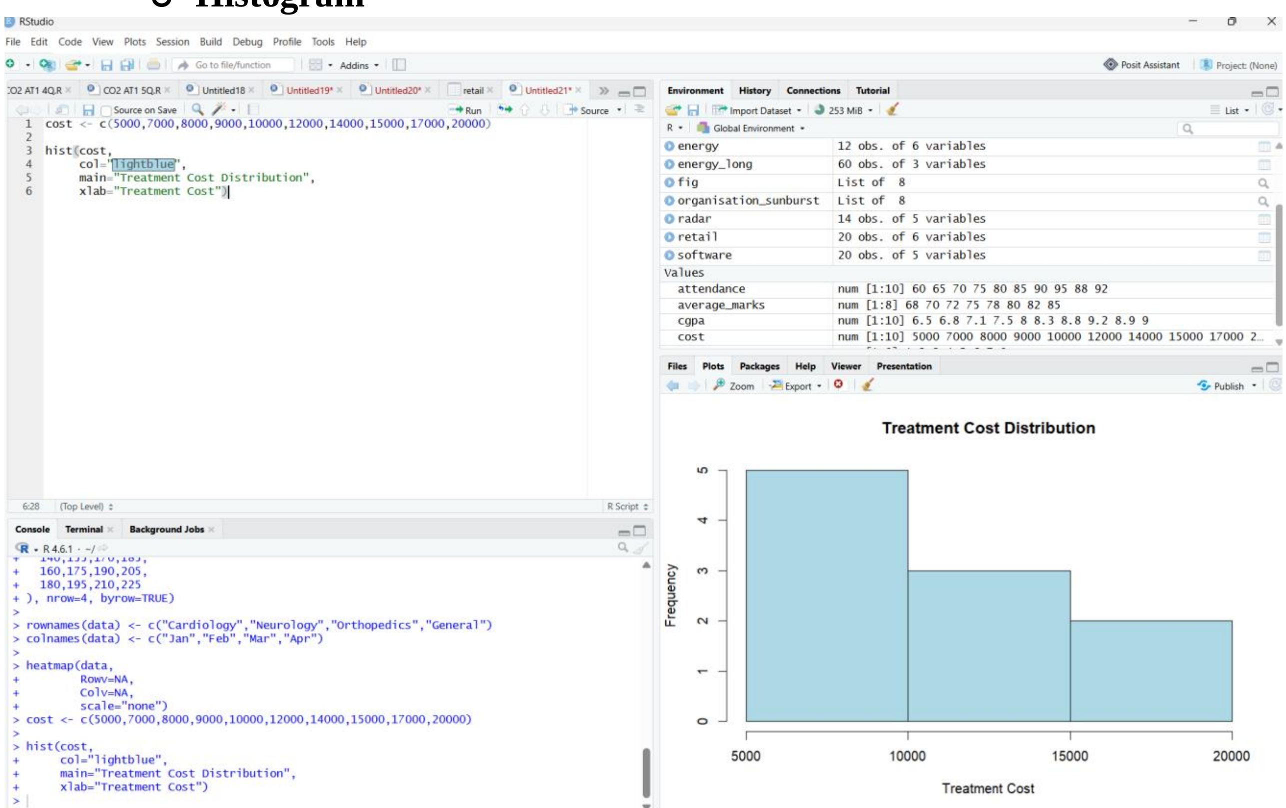
Answer:

Requirement	**Suitable Visualization**	**Justification**
Monthly patient admissions	**Heatmap**	Shows admission patterns across different months effectively.
Distribution of treatment costs	**Histogram**	Displays the frequency distribution of treatment costs.
Disease category proportions	**Violin Plot**	Compares the distribution of different disease categories.
Age vs Recovery Time	**Scatter Plot**	Shows the relationship between patient age and recovery time.
Department-wise treatment costs	**Density Plot**	Displays the distribution of treatment costs for departments.

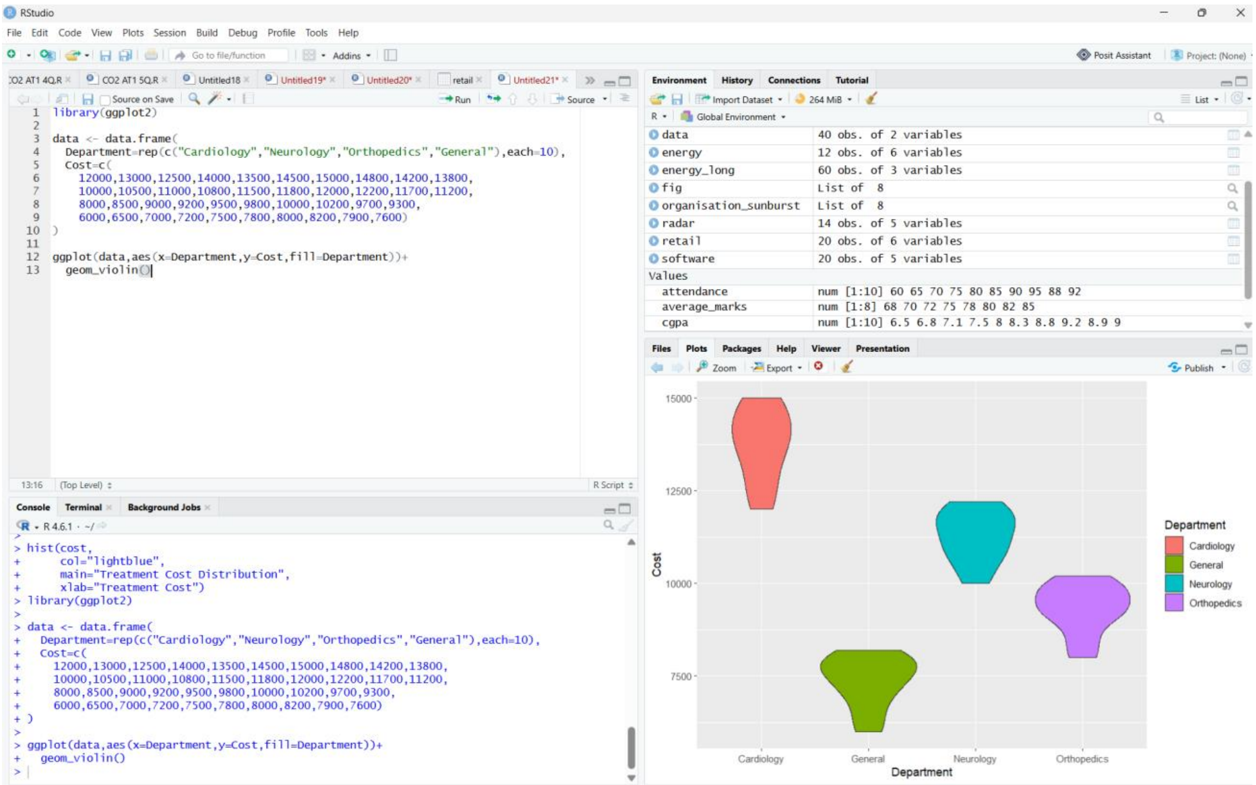
2. Create in **R**:
    - **Heatmap**


data <- matrix(c(
120,135,150,165,
140,155,170,185,
160,175,190,205,
180,195,210,225
), nrow=4, byrow=TRUE)
rownames(data) <- c("Cardiology","Neurology","Orthopedics","General")
colnames(data) <- c("Jan","Feb","Mar","Apr")
heatmap(data,
Rowv=NA,
Colv=NA,
scale="none")
General
Orthopedics
Neurology
Cardiology
Jan
Feb
Mar
Apr

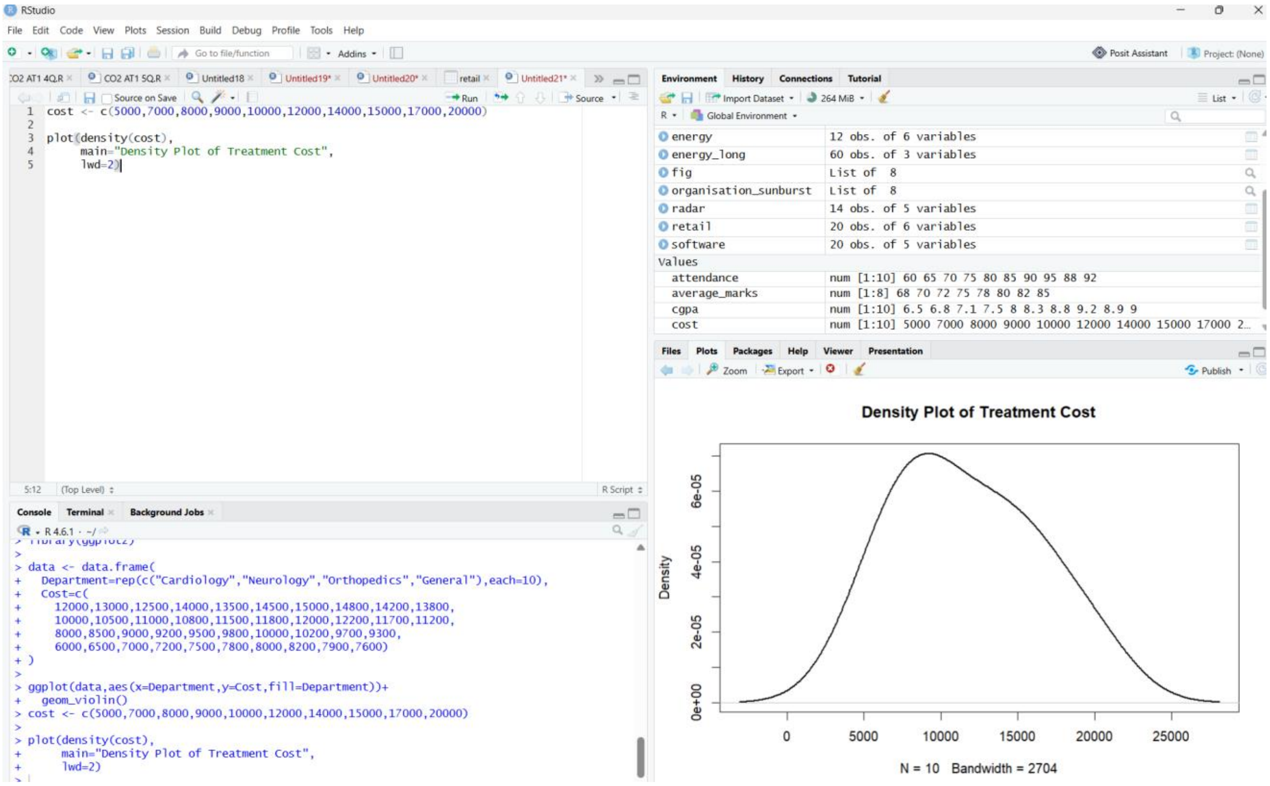
- **Histogram**


cost <- c(5000,7000,8000,9000,10000,12000,14000,15000,17000,20000)
hist(cost,
col="lightblue",
main="Treatment Cost Distribution",
xlab="Treatment Cost")
Treatment Cost Distribution
Frequency
Treatment Cost
5000
10000
15000
20000

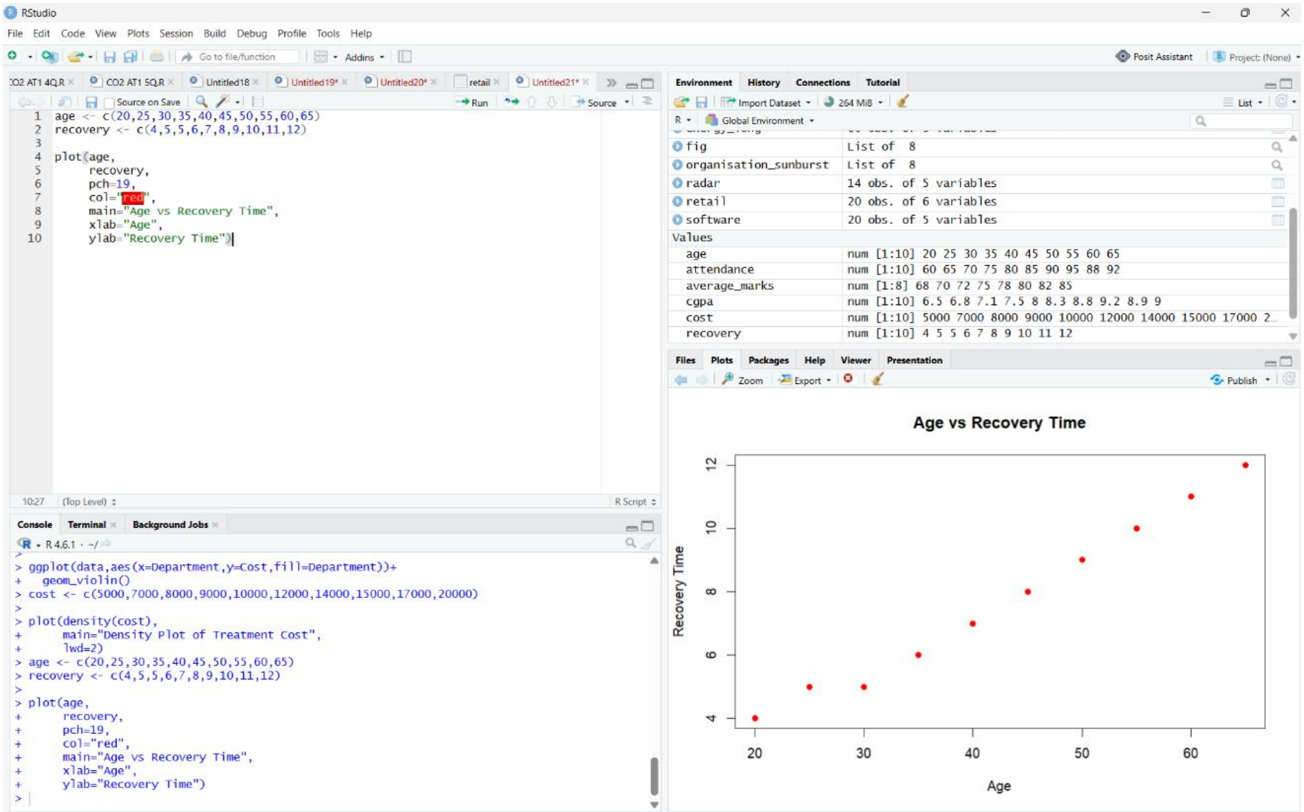
- ## Violin Plot



- ## Density Plot



- **Scatter Plot**



3. Interpret:

○ Which department has the highest treatment cost?

Answer:

The **Cardiology** department has the highest treatment cost among all departments.

○ Which disease category occupies the largest proportion?

Answer:

The Cardiovascular disease category occupies the largest proportion of patients.

○ Is patient age associated with recovery time?

Answer:

Yes. The scatter plot indicates a positive relationship, meaning that as patient age increases, recovery time also tends to increase.

Case Study 4 – Smart City Traffic Monitoring

Scenario

A smart city collects traffic sensor data from different junctions. The dataset contains: • Junction

• Time of day

• Vehicle count

• Vehicle type

• Average speed

• GPS coordinates

City planners aim to reduce traffic congestion.

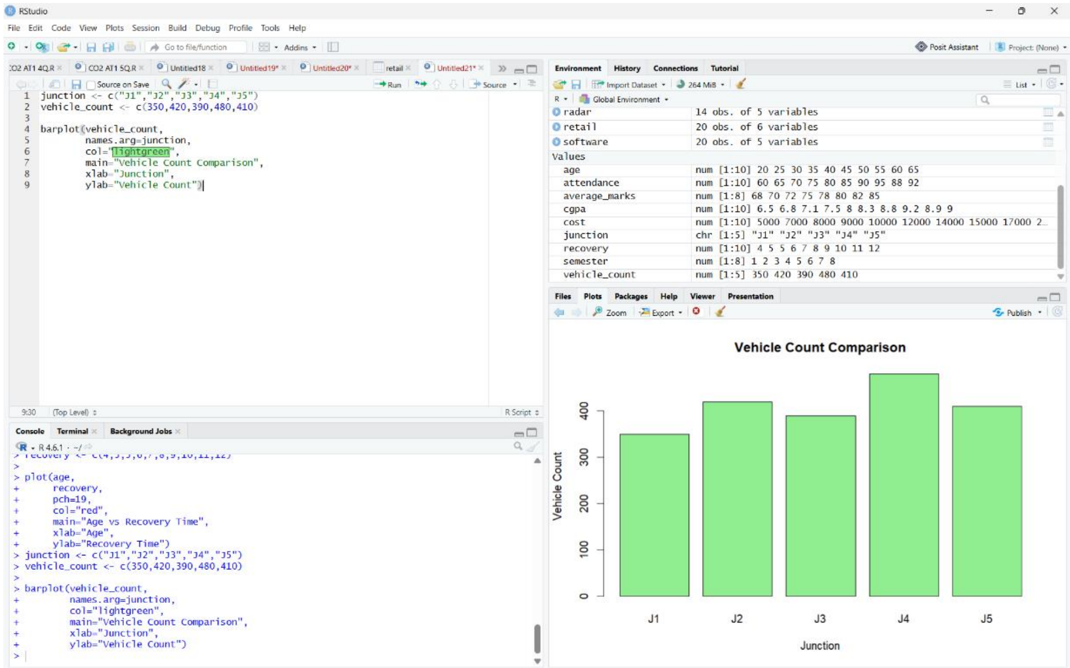
Tasks

1. Select suitable visualizations for:
   - Vehicle count comparison
   - Hourly traffic trend
   - Vehicle type proportion
   - Traffic density by location
   - Relationship between traffic volume and average speed

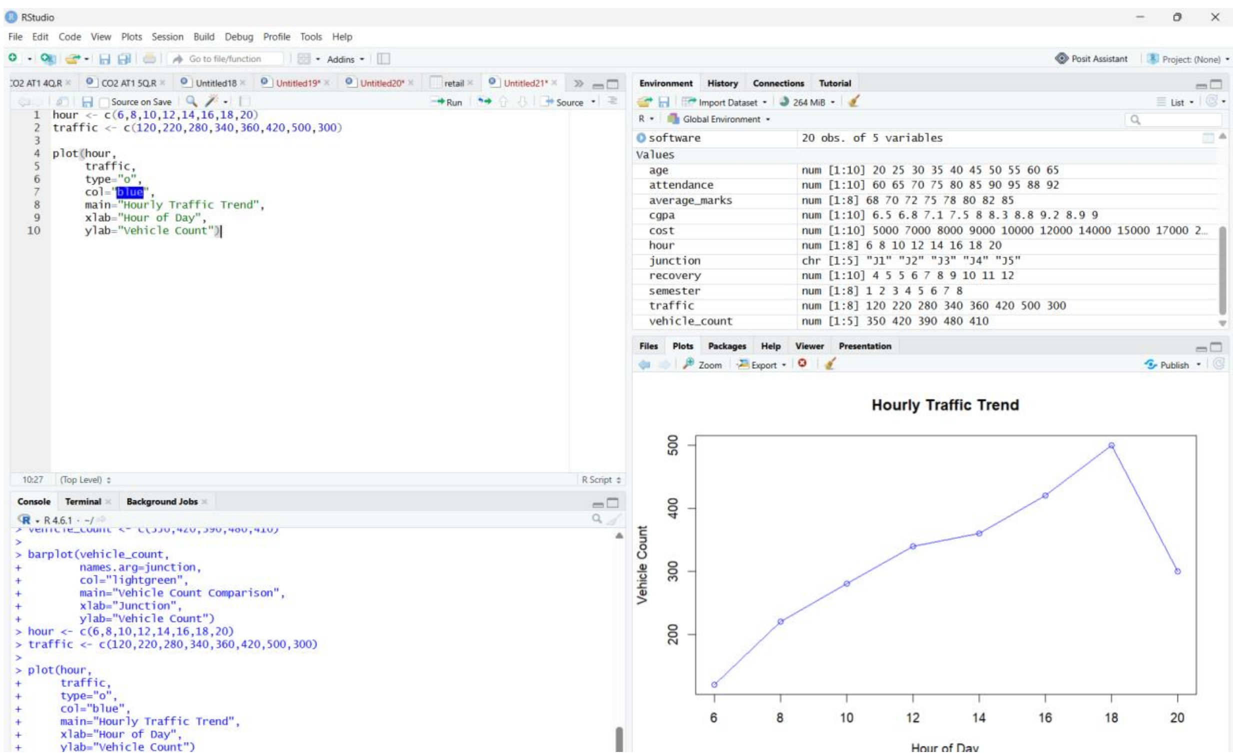
Answer:

Requirement	**Suitable Visualization**	**Justification**
Vehicle count comparison	**Bar Plot**	Compares the number of vehicles at different junctions.
Hourly traffic trend	**Line Chart**	Shows traffic flow changes throughout the day.
Vehicle type proportion	**Pie Chart**	Displays the percentage of each vehicle type.
Traffic density by location	**Geospatial Map**	Represents traffic density across different locations.
Relationship between traffic volume and average speed	**Scatter Plot**	Shows the relationship between traffic volume and vehicle speed.

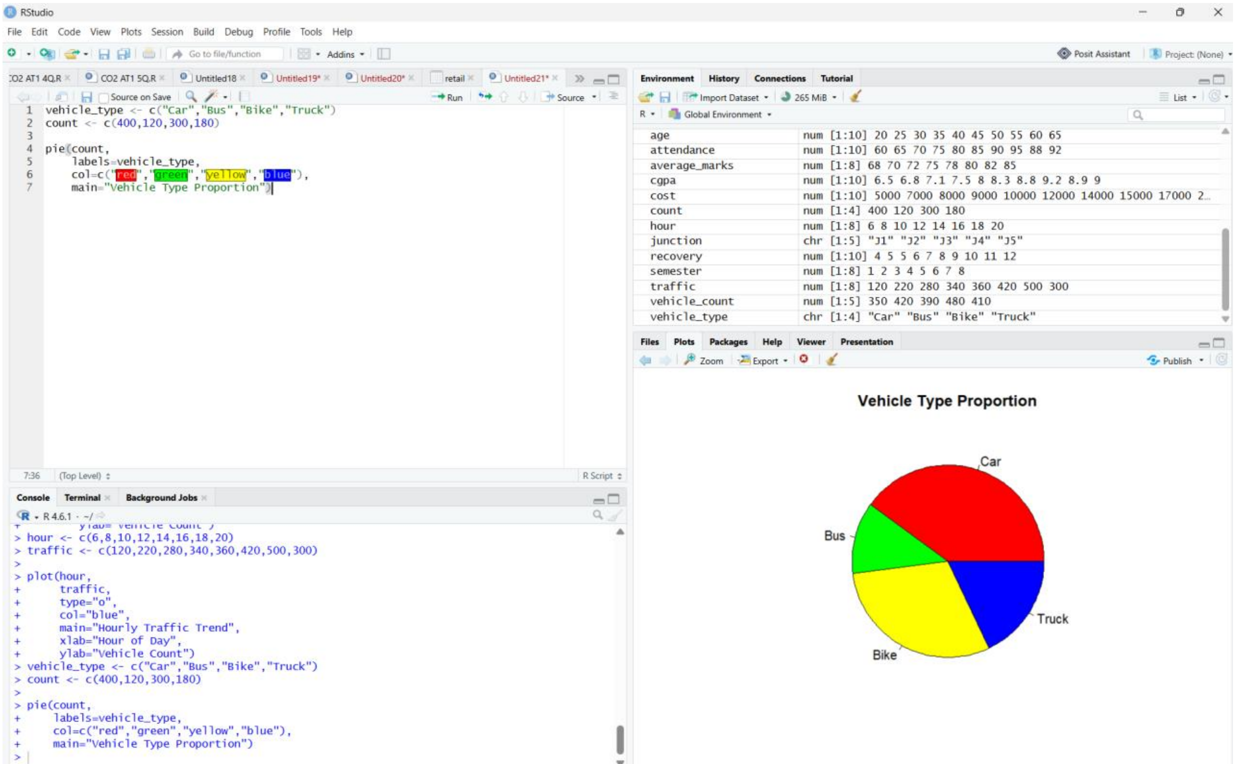
2. Using **R**, create:
   - **Bar Plot**



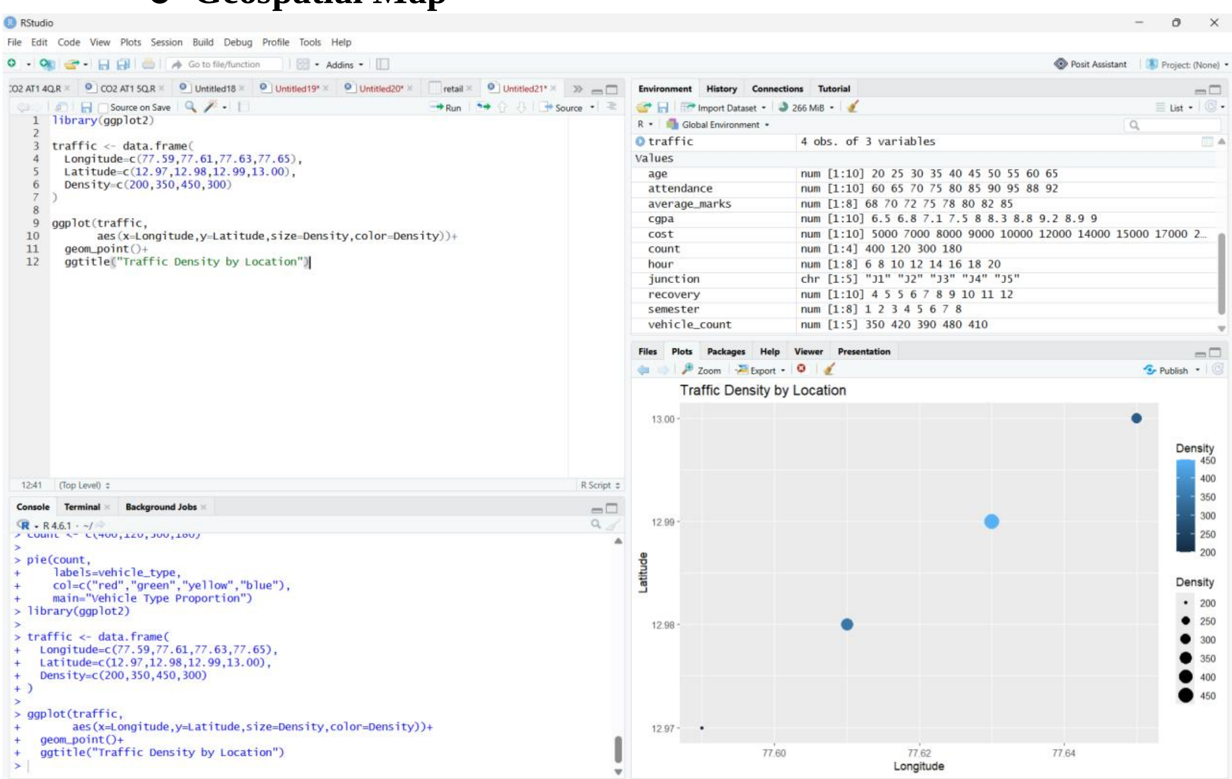
- **Line Chart**



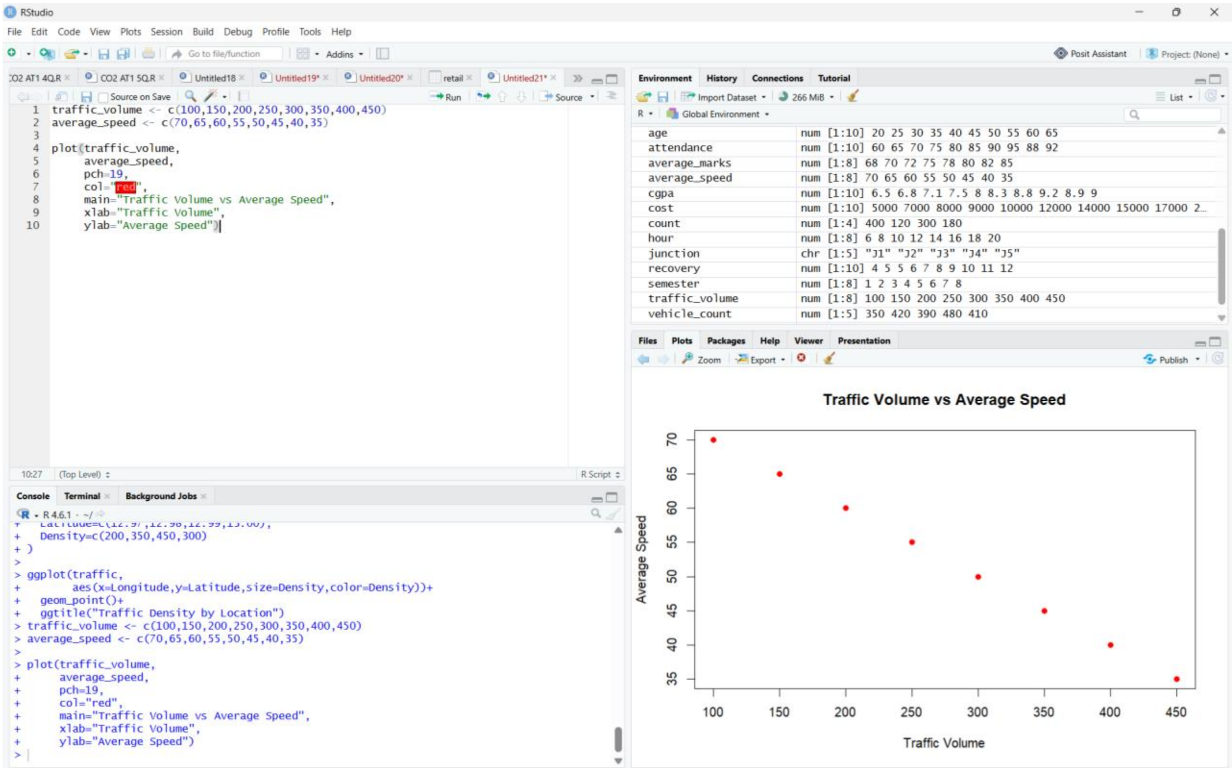
- **Pie Chart**



- **Geospatial Map**



- **Scatter Plot**



3. Interpret:

○ Which junction experiences the highest congestion?

Answer:

J4 experiences the highest congestion because it has the highest vehicle count.

○ During which hours is traffic highest?

Answer:

Traffic is highest during the evening peak hours (around 6 PM), where the vehicle count reaches its maximum.

○ Does increased traffic reduce vehicle speed?

Answer:

Yes. The scatter plot shows a **negative relationship**, indicating that as traffic volume increases, the average vehicle speed decreases.

Case Study 5 – Social Media User Interaction Analysis

Scenario

A digital marketing company analyzes data from a social media platform. The dataset includes: • User ID

• Followers

• Likes

• Shares

• Comments

• Posting time

• User interactions

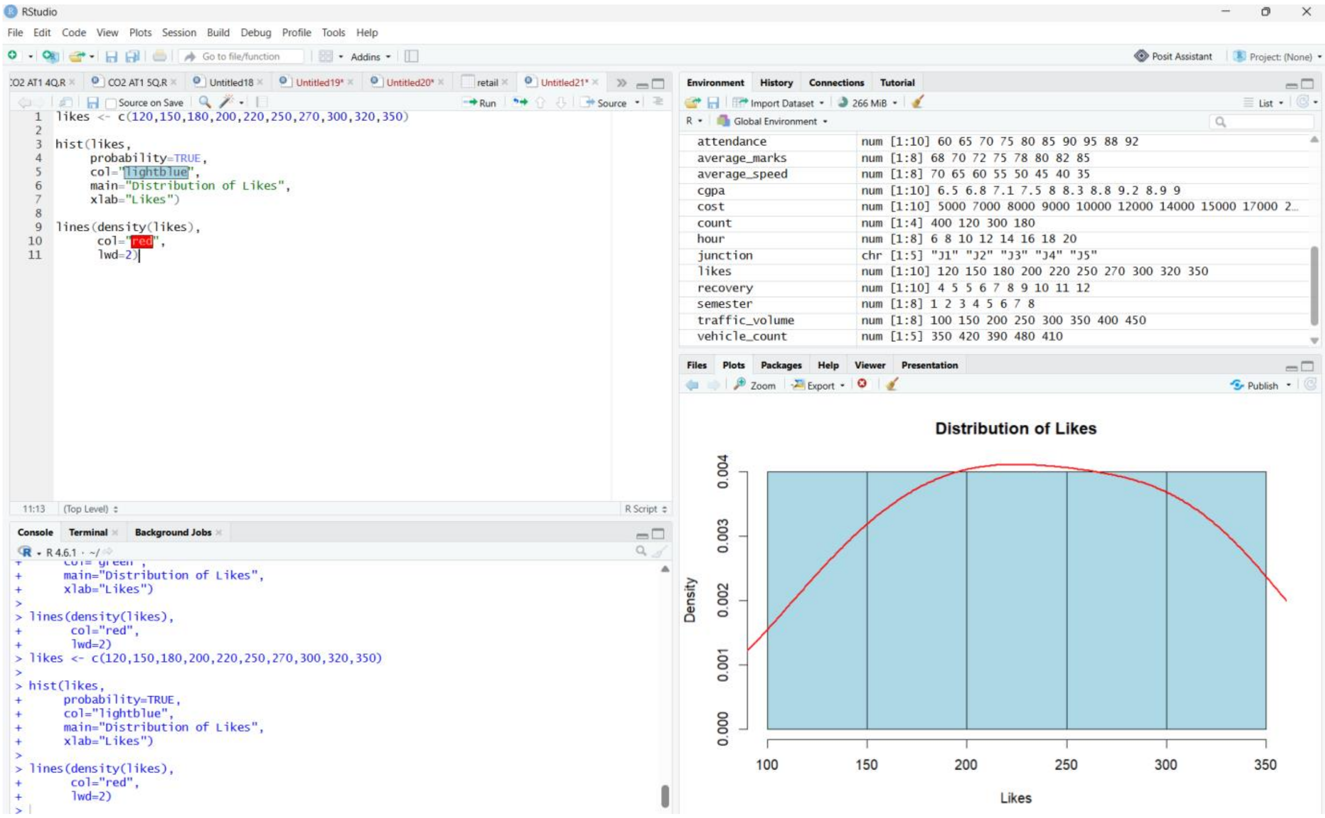
The company wants to understand engagement patterns and user networks. **Tasks**

1. Select appropriate visualizations for:
    - Distribution of likes
    - Relationship between followers and engagement
    - User interaction network
    - Posting activity over time
    - Multivariate comparison of likes, shares, and comments

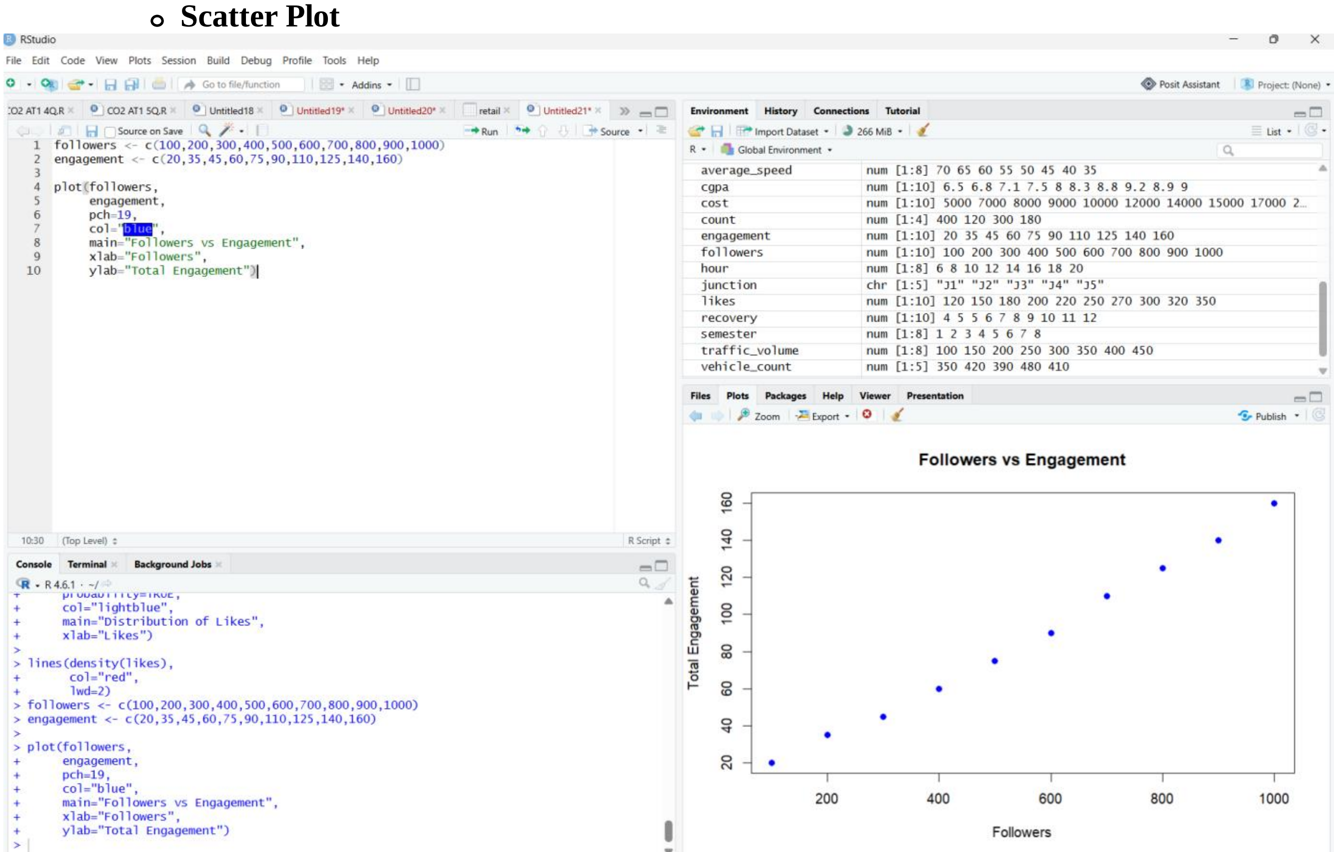
Answer:

Requirement	Suitable Visualization	Justification
Distribution of likes	**Histogram with Density Curve**	Shows the frequency and distribution pattern of likes.
Relationship between followers and engagement	**Scatter Plot**	Identifies the relationship between followers and engagement.
User interaction network	**Network Visualization**	Displays connections and interactions among users.
Posting activity over time	**Temporal Line Plot**	Shows changes in posting activity over time.
Multivariate comparison of likes, shares, and comments	**Heatmap**	Compares multiple engagement metrics simultaneously.

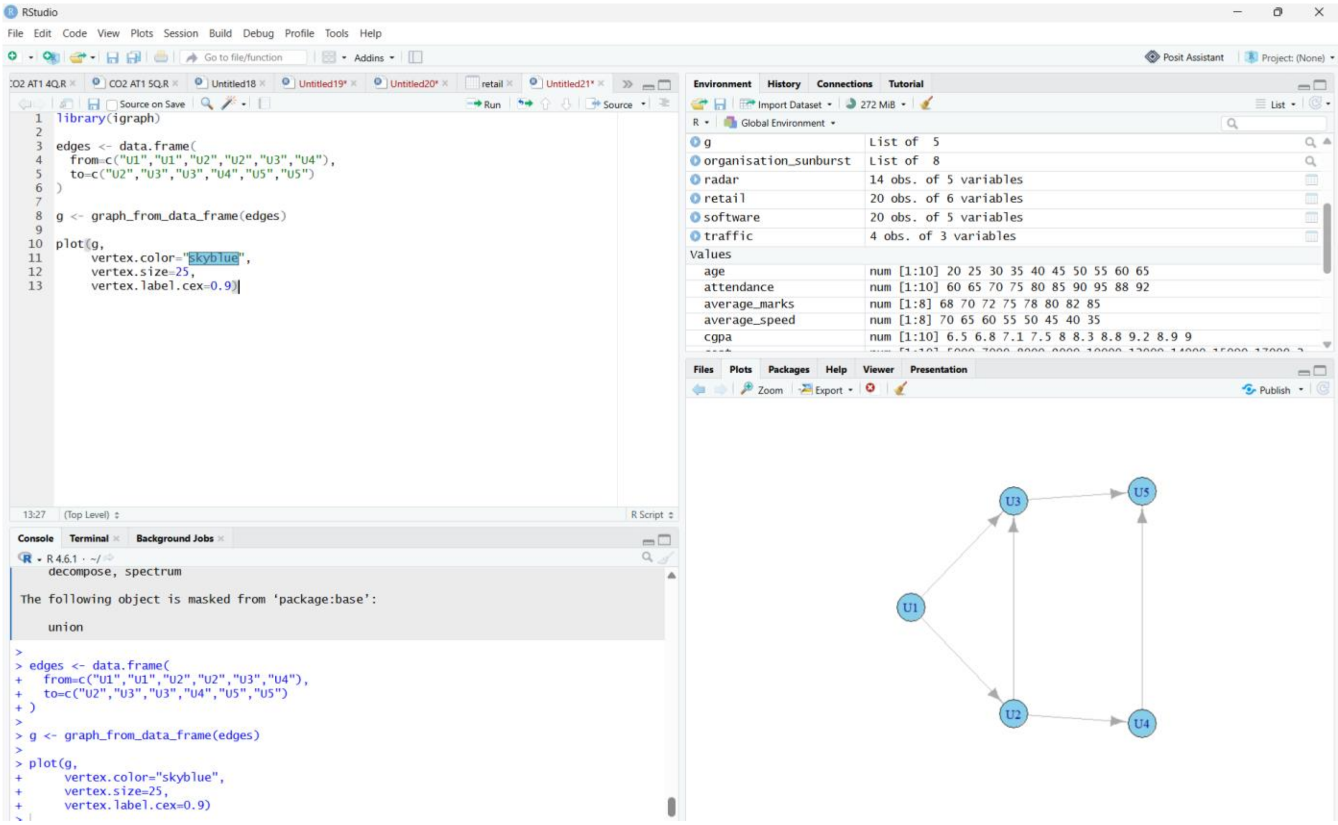
2. Develop in **R**:
    - **Histogram with Density Curve**



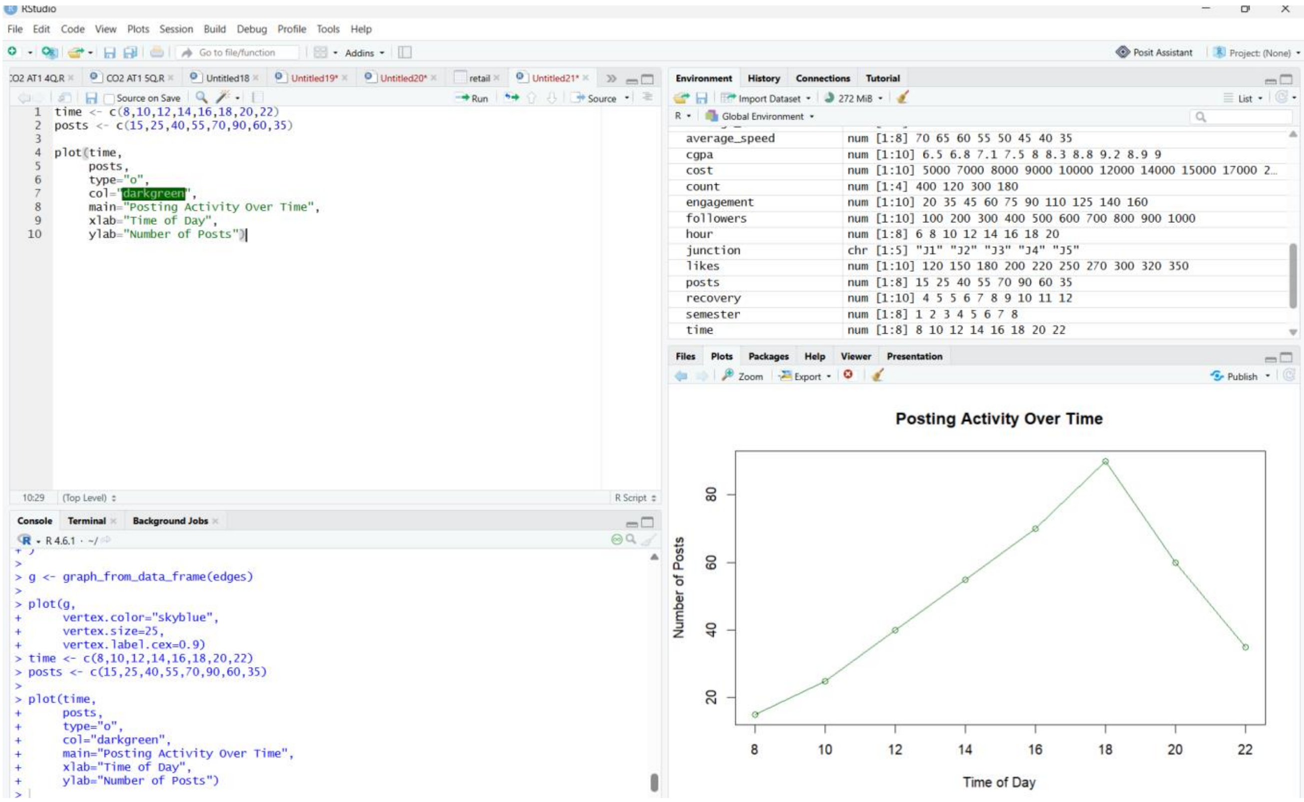
- **Scatter Plot**



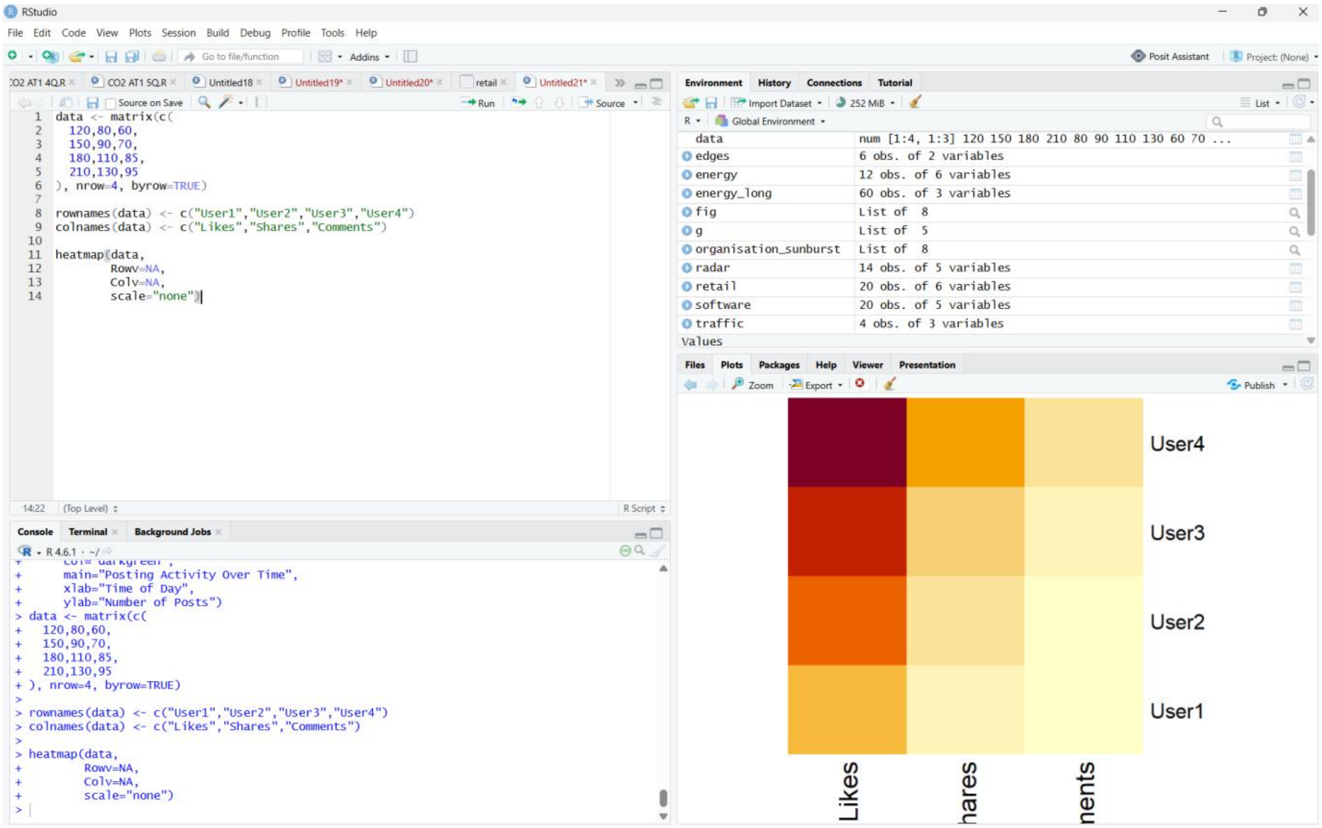
- ## Network Visualization



- ## Temporal Line Plot



- **Heatmap**



3. Interpret:

○ Which users are the most influential?

Answer:

Users with the highest number of followers and engagement are the most influential because they receive more likes, shares, and comments.

○ At what time is engagement highest?

Answer:

Engagement is highest during the evening (around 6 PM), when the number of user interactions reaches its maximum.

○ Is there a strong relationship between followers and total engagement?

Answer:

Yes. The scatter plot shows a strong positive relationship, indicating that users with more followers generally receive higher engagement.

○ Identify highly connected users from the network visualization.

Answer:

U2 and U3 are the most highly connected users because they have the greatest number of connections with other users.

○ Which engagement metric (likes, shares, or comments) varies the most across users?

Answer:

Likes show the greatest variation across users, followed by shares and comments.